

Model Explainability & Interpretability

Summary Sheet

An abstract geometric pattern consisting of teal lines and dots on a dark background. The pattern forms a complex, interconnected structure that resembles a network or a stylized architectural design. The lines are of varying lengths and thicknesses, and the dots are small, solid teal circles. The overall effect is a modern, tech-inspired aesthetic.

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Summary Sheet

Learning objectives - Through this module you will learn....

- What interpretable models are and how they differ from explainable models.
- What drives the necessity to make models interpretable and/or explainable and who the key stakeholders are.
- Where and how to ensure model interpretability in the ML pipeline
- Techniques to make a model interpretable or explainable both at the global level (i.e. general behavior of the model) and the local level (i.e. reliance of the model on particular features for a particular observation)
- How the global explanation techniques of Permutation Feature Importance and SHAP work.
- How the local explanation techniques of LIME and ANCHOR work.

Overview

This module dives deep into why, when, and where the decisions of the predictive models need to be explained. With practical examples and use cases, you will learn to appreciate how the predictive power of the machine learning models is not the only goal of a data scientist. Depending on the application, model interpretability and explainability play a huge role in the model choice. You will learn about well-researched techniques to render the black-box machine learning models with explainability.

Why this topic matters

The use of machine learning models is pervasive in our daily lives, ranging from online recommendations and voice recognition systems on our phones to making loan decisions. Naturally, we want these models to be as accurate as possible. However, this comes at the cost of making the models more complex, making them harder to interpret. In the absence of visibility into these black-box models, we risk incorporating the systemic biases (e.g., racial/gender discrimination) into the models' decision-making ability. Therefore, to deploy these models in the real world, it is vital to either make the models interpretable from the beginning or, if the models are very complex, use relevant techniques to extract explanations for their decisions.

Practical use cases and applications

In most safety-critical applications such as cancer diagnosis or autonomous driving, humans must understand the model behavior. For example, in cancer treatment, a predictive model might suggest pursuing a treatment plan with severe side effects. The clinician would require an explanation/rationalisation that will convince them that the recommendation is sensible.

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This article from [Reuters](#) explains how Amazon's ML experts discovered that their predictive models for recruiting candidates relied on gender discriminatory factors. Several other corporations are turning to AI for recruiting automation, and the article lists several of them.

This blog from [Harvard University](#) delves deep into various forms of inequities prevalent in the current facial recognition systems, which are being used everywhere, ranging from law enforcement schemes to housing decisions. How different corporations like IBM and Microsoft have taken steps to reduce such biases are also discussed.

How Model Explainability & Interpretability creates value and can be useful to organisations

Predictive models are increasingly being used in the value chain of several organizations. While the goal is to design models with superior performance, it comes at the cost of making the models more complex, making it difficult to reason why the model behaves in the way it does.

On the other hand, a model you can interpret and understand is one you can more easily improve, thereby delivering new kinds of intelligent products. Besides, in most organizations, applications of models are constrained by legal regulations and ethical concerns, making model interpretability of utmost importance. Thus, with the help of the tools and techniques to explain model predictions, data scientists are better positioned to deliver more reliable models that can be safely deployed in the real world.

Module Components - Time & Difficulty

LU/PAK	Est. Time to complete	Difficulty Level
L01 Introduction to Interpretability and Explainability	2hr	Easy
L02 Global Explanations	3hr	Medium
L03 Local Explanations	3hr	Medium
L04 Explanation methods for deep neural networks	2hr	Hard e-learning
PAK1: Lime & Anchor	2hr	Medium

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Tools, technique, libraries covered in this module

This module is taught entirely in Python. We will use multiple machine learning libraries:

- Pandas
- Numpy
- Scikit-learn
- LightGBM
- Anchor
- Lime
- Shap

Which topics we will cover, and why

This course dives deep into why machine learning models need to be interpretable. It will provide the learner with the tools and in-depth understanding of explainability techniques for machine learning models related explicitly to tabular and text data. The learner will use these techniques to design practically deployable trustworthy models.

Prerequisite knowledge needed to be able to work with this module

- Basic data analysis
- Basic notion of linear algebra
- Basic understanding of classification and regression methods like logistic regression, decision trees, etc.
- Basic understanding of Neural Networks

Which other topics are NOT encompassed in this module

This module does not cover recent advances in interpretability and explainability of black-box models for more diverse applications such as computer vision or voice recognition systems.

Model Explainability & Interpretability Summary Sheet

Suggested additional readings

Rudin et al. (2021) Interpretable Machine Learning: Fundamental Principles and 10 Grand Challenges

Christoph Molnar 2020, [Interpretable Machine Learning](#)

Arrieta et al. (2019) Explainable Artificial Intelligence (XAI): Concepts, Taxonomies, Opportunities and Challenges toward Responsible AI

Preece et al. (2018) Stakeholders in Explainable AI



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